Stockholm University

Department of Meteorology

Numerical Methods in Atmospheric Sciences and Oceanography (MO8007)

Home Assignment 2

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Question

Exercise 1 – Staggered vs Un-staggered Grid

The following code lines are taken from a larger FORTRAN code that interpolates the temperature at a given position (described by a real pseudo-index x_u). The original code works for an Arakawa C-grid. For simplicity we will consider the 1-D case where the Arakawa grid is translated to a staggered grid. The (x_u) is defined using u-points as a reference while the temperature is defined in an h-point.

```
SUBROUTINE interp(xu,T)

    ! computes temperature at the position given by xu by interpolating data
    IMPLICIT NONE
    INTEGER :: ip, im
    REAL :: xu, ax, Tint
    REAL, DIMENSION(:), INTENT(IN) :: T
    [ ... ]
    ip = NINT(xu) + 1
    im = NINT(xu)
    ax = REAL(ip) - xu
    Tint = T(im)*ax + T(ip)*(1-ax)
    [ ... ]

END SUBROUTINE
```

Consider the case $x_u = 10.4$ and the temperature values: $T(9) = 5$, $T(10) = 10$, $T(11) = 20$.

(a) Using the code above compute the interpolated temperature (T_{int}) at x_u.

(b) The obtained result for interpolated temperature (T_{int}) is wrong. Why? Fix the code (explain the changes) and recalculate T_{int}.

(c) Would the original interpolation scheme work correctly (for any x_u value) if an unstaggered grid was used instead?

Exercise 2 – Order of Accuracy

Consider the continuity equation of the 1-D shallow water equations:

$$\frac{\partial h}{\partial t} = -H\frac{\partial u}{\partial x}.$$

(a) Derive the order of accuracy for the two differences of the discretised equations centred in time and using an unstaggered grid.

(b) Repeat the previous task using instead a staggered grid.

1 First exercise

(a) Initially the code had a problem that it uses the `NINT` function and not the `FLOOR` function. This gives incorrect interpolated values when x_u has a decimal part above 0.5. The code was thus changed to

```
SUBROUTINE interp(xu, T, Tint)
    IMPLICIT NONE
    REAL, INTENT(IN) :: xu
    REAL, DIMENSION(:), INTENT(IN) :: T
    REAL, INTENT(OUT) :: Tint
    REAL :: ax
    INTEGER :: ip, im

    ip = FLOOR(xu) + 1
    im = FLOOR(xu)
    ax = REAL(ip) - xu

    Tint = T(im)*ax + T(ip)*(1 - ax)
```

Using this corrected code the interpolated temperature at position $x_u = 10.4$ was calculated to be $T_{int} = 14$.

(b) However, the code provided had one more issue, since does not use a staggered grid. If we assume a staggered grid then the temperature points a eat the h-points not the u-points(x_u). are not at the same positions but of set by Δx. In the staggered grid the we have that $h_j = u_{j-\frac{1}{2}}$. Another way of stating this is that the point u_j is surrounded by

$$h_j < u_j < u_{j+1}. \tag{1}$$

However, one must note that the staggering can either be forward or backward, however the common way is to use what i showed above. One can thus modify the code so that the position $x_u = 10.4$ lies in the position $x_h = 9.9$. One thus has the code as:

```
SUBROUTINE interp(xu, T, Tint)
    IMPLICIT NONE
    REAL, INTENT(IN) :: xu
    REAL, DIMENSION(:), INTENT(IN) :: T
    REAL, INTENT(OUT) :: Tint
    REAL :: ax, xh
    INTEGER :: ip, im

    xh = xu-0.5 ! Here we change to h-points

    ip = FLOOR(xh) + 1
    im = FLOOR(xh)
    ax = REAL(ip) - xh

    Tint = T(im)*ax + T(ip)*(1 - ax)
```

Answer

One thus obtains an answer of $T(x_u = 10.4) = T(x_h = 9.9) = 9.5$.

(c) If an unstaggered grid was used then the original code would work just fine since this would imply that $x_u = x_h$.

2 Second exercise

(a) The continuity equation is

$$\frac{\partial h}{\partial t} = -H\frac{\partial u}{\partial x}, \tag{2}$$

where $h(x,t)$ is the height of the water and $u(x,t)$ is the zonal velocity of the fluid. Discretizing this equation cantered in time and space yields on an unstaggered grid:

$$\frac{h_j^{n+1} - h_j^{n-1}}{2\Delta t} = -H\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x}. \tag{3}$$

Using Taylor expansions as

$$h_j^{n\pm1} = h_j^n \pm \Delta t\left(\frac{\partial h}{\partial t}\right)^n + \frac{[\Delta t]^2}{2}\left(\frac{\partial^2 h}{\partial t^2}\right)^n \pm \frac{[\Delta t]^3}{6}\left(\frac{\partial^3 h}{\partial t^3}\right)^n + \mathcal{O}([t^4]), \tag{4}$$

$$u_{j\pm1}^n = u_j^n \pm \Delta x\left(\frac{\partial u}{\partial x}\right)_j + \frac{[\Delta x]^2}{2}\left(\frac{\partial^2 u}{\partial x^2}\right)_j \pm \frac{[\Delta x]^3}{6}\left(\frac{\partial^3 u}{\partial x^3}\right)_j + \mathcal{O}([x^4]), \tag{5}$$

one yields

$$\frac{h_j^{n+1} - h_j^{n-1}}{2\Delta t} = \frac{\partial h}{\partial t} + \frac{[\Delta t]^2}{6}\left(\frac{\partial^3 h}{\partial t^3}\right)^n + \mathcal{O}([t^3]), \quad \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{[\Delta x]^2}{6}\left(\frac{\partial^3 u}{\partial x^3}\right)_j + \mathcal{O}([x^3]). \tag{6}$$

Thus, the scheme has an accuracy of second-order in both time and space.

(b) If one uses a staggered grid on Equation 2, one obtains the discretisation:

$$\frac{{h_{j_h}}^{n+1} - h_{j_h}^{n-1}}{2\Delta t} = -H\frac{u_{j_u}^n - u_{j_u-1}^n}{\Delta x}, \tag{7}$$

where one has to remember that $x_u = j\Delta x$ while $x_h = x_u - \frac{1}{2}\Delta x$. Using the Taylor expansions, where the time derivative of the h-points behave exactly as before. However the spatial derivative of the u-points, one now has to remember is actually centred, since $x_h = x_u(j - \frac{1}{2})$.

We thus expand u_j^n and u_{j-1}^n around the point $x_h = \left(j - \frac{1}{2}\right)\Delta x$:

$$u_j^n = u_{j_h+\frac{1}{2}}^n \tag{8}$$

$$= u_{j_h}^n + \frac{\Delta x}{2}\left(\frac{\partial u}{\partial x}\right)_{j_h} + \frac{[\Delta x]^2}{8}\left(\frac{\partial^2 u}{\partial x^2}\right)_{j_h} + \frac{[\Delta x]^3}{48}\left(\frac{\partial^3 u}{\partial x^3}\right)_{j_h} + \mathcal{O}([x^4]), \tag{9}$$

$$u_{j-1}^n = u_{j_h-\frac{1}{2}}^n \tag{10}$$

$$= u_{j_h}^n - \frac{\Delta x}{2}\left(\frac{\partial u}{\partial x}\right)_{j_h} + \frac{[\Delta x]^2}{8}\left(\frac{\partial^2 u}{\partial x^2}\right)_{j_h} - \frac{[\Delta x]^3}{48}\left(\frac{\partial^3 u}{\partial x^3}\right)_{j_h} + \mathcal{O}(\Delta\left[x^4\right]). \tag{11}$$

Subtracting the two expressions and dividing by Δx yields:

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \left(\frac{\partial u}{\partial x}\right)_{j_h} + \frac{[\Delta x]^2}{24}\left(\frac{\partial^3 u}{\partial x^3}\right)_{j_h} + \mathcal{O}(\Delta\left[x^3\right]). \tag{12}$$

Thus, this scheme also has an accuracy of second-order in both time and space.

Answer

Thus, the order of accuracy in the staggered and unstaggered grid is of the second-order in both time and space, $\mathcal{O}([x^2])$ and $\mathcal{O}([t^2])$.

However the new prefactor in the spatial accuracy is $\frac{1}{24}$ compared to the old $\frac{1}{6}$, and is thus a quarter as big as in the unstaggered case. Thus, the staggered scheme has a smaller truncation error than the unstaggered scheme and will yield spatially more precise result.